

1. Wap in 'Python' language to accept two values and calculate highest value and lowest value.

```
a=int(input("Enter a 1 no.: "))
b=int(input("Enter a 2 no.: "))
```

```
if a > b:
```

```
    print("Highest value is:", a)
```

```
    print("Lowest value is:", b)
```

```
elif b > a:
```

```
    print("Highest value is:", b)
```

```
    print("Lowest value is:", a)
```

```
else:
```

```
    print("Both numbers are equal.")
```

2. Wap in 'Python' language to accept a number, check and display the message whether it is positive

or negative number.

```
a=int(input("Enter a 1 no.: "))
```

```
if a>0:
```

```
    print("positive Number")
```

```
elif a<0:
```

```
    print("Negative Number")
```

```
else:
```

```
    print("Number is Zero")
```

3. Wap in 'Python' language to accept a number, check and display the message whether it is even or

odd number.

```
a=int(input("Enter a 1 no.: "))
```

```
if a%2==0:
```

```
    print("even Number")
```

```
else:
```

```
    print("Odd Number")
```

4. Wap in 'Python' language to accept a number, check and display the message whether it is positive
even or positive odd or negative even or negative odd.

```
a=int(input("Enter a 1 no.: "))
```

```
if a>0:
```

```
    if a%2==0:
```

```
        print("No is positve even")
```

```
    else:
```

```
        print("Positive Odd")
```

```
elif a<0:
```

```
    if a%2==0:
```

```
        print("Negative even")
```

```
    else:
```

```
        print("Negative odd")
```

```
else:
```

```
    print("No is ZERO")
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: 20
Enter a 2 no.: 10
Highest value is: 20
Lowest value is: 10
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: 10
Enter a 2 no.: 10
Both numbers are equal.
PS C:\Users\DELL\OneDrive\Desktop\python> █
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> pyth
Enter a 1 no.: 10
positive Number
PS C:\Users\DELL\OneDrive\Desktop\python> pyth
Enter a 1 no.: -10
Negative Number
PS C:\Users\DELL\OneDrive\Desktop\python> pyth
Enter a 1 no.: 0
Number is Zero
PS C:\Users\DELL\OneDrive\Desktop\python> █
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: 52
even Number
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: 5
Odd Number
PS C:\Users\DELL\OneDrive\Desktop\python> █
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\U
Enter a 1 no.: 12
No is positve even
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\U
Enter a 1 no.: 33
Positive Odd
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\U
Enter a 1 no.: -33
Negative odd
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\U
Enter a 1 no.: -12
Negative even
PS C:\Users\DELL\OneDrive\Desktop\python> █
```

5. Wap in 'Python' language to accept a character, check and display the message whether it is a digit

or an alphabet or a special symbol.

```
ch=ord(input("Enter a character:"))
```

```
if ch>=65 and ch<=90:
```

```
    print("An Alphabet")
```

```
elif ch>=97 and ch<=122:
```

```
    print("An Alphabet")
```

```
elif ch>=48 and ch<=57:
```

```
    print("Digits")
```

```
else:
```

```
    print("Special Symbol")
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c
Enter a character:A
An Alphabet
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c
Enter a character:j
An Alphabet
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c
Enter a character:5
Digits
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c
Enter a character:@
Special Symbol
PS C:\Users\DELL\OneDrive\Desktop\python> █
```

#6 Wap in 'Python' to accept a character, and determine whether the character is small case or capital case

letter.

```
ch=ord(input("Enter a character:"))
```

```
if ch>=65 and ch<=90:
```

```
    print("Capital letter")
```

```
elif ch>=97 and ch<=122:
```

```
    print("Small letter")
```

```
elif ch>=48 and ch<=57:
```

```
    print("Digits")
```

```
else:
```

```
    print("Otherwise")
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a character:h
Small letter
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a character:A
Capital letter
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a character:
Otherwise
PS C:\Users\DELL\OneDrive\Desktop\python> █
```

7. Wap in 'Python' language to accept a number, check and display the message whether it is positive #or

negative or neither +ve nor -ve.

```
a=int(input("Enter a 1 no.: "))
```

```
if a>0:
```

```
    print("positive Number")
```

```
elif a<0:
```

```
    print("Negative Number")
```

```
else:
```

```
    print("neither +ve nor -ve")
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: 52
positive Number
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: 669
positive Number
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: -2
Negative Number
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter a 1 no.: 0
neither +ve nor -ve
PS C:\Users\DELL\OneDrive\Desktop\python> █
```

8. Wap in 'Python' to accept total no. in 500 and find out percentage and display remarks, where remarks

will be given sequence and format:-

per >= 30 and per < 45 pass

per >= 45 and per < 60 second

per >= 60 and per <= 100 first

otherwise fail.

total = 500

M = int(input("Enter total marks obtained (out of 500): "))

per = M / total * 100

print("Percentage =", per, "%")

if per >= 30 and per < 45:

 print("Remark: Pass")

elif per >= 45 and per < 60:

 print("Remark: Second Division")

elif per >= 60 and per <= 100:

 print("Remark: First Division")

else:

 print("Remark: Fail")

```
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\8.py"
Enter total marks obtained (out of 500): 385
Percentage = 77.0 %
Remark: First Division
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\8.py"
Enter total marks obtained (out of 500): 255
Percentage = 51.0 %
Remark: Second Division
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\8.py"
Enter total marks obtained (out of 500): 200
Percentage = 40.0 %
Remark: Pass
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\8.py"
Enter total marks obtained (out of 500): 20
Percentage = 4.0 %
Remark: Fail
PS C:\Users\DELL\OneDrive\Desktop\python>
```

9. Wap in 'Python' to accept year and display a message whether the year is a leap year or not.

year = int(input("Enter a year: "))

if year % 100 == 0:

 if year % 400 == 0:

 print("Given year", year, "is a leap year")

 else:

 print("Given year", year, "is not a leap year")

elif year % 4 == 0:

 print("Given year", year, "is a leap year")

else:

 print("Given year", year, "is not a leap year")

```
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\9.py"
Enter a year: 2004
Given year 2004 is a leap year
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\9.py"
Enter a year: 2025
Given year 2025 is not a leap year
PS C:\Users\DELL\OneDrive\Desktop\python>
```

10. Wap in 'Python' to accept six digit no. from keyboard, check and display a message whether given

number is palindrome or not.

n=int(input("Enter a 6-Digit no. : "))

temp = n

d1=n%10

n=n//10

d2=n%10

n=n//10

d3=n%10

n=n//10

d4=n%10

n=n//10

d5=n%10

n=n//10

rev=d1*100000+d2*10000+d3*1000+d4*100+d5*10+n

if temp==rev:

 print("Palindrome")

else:

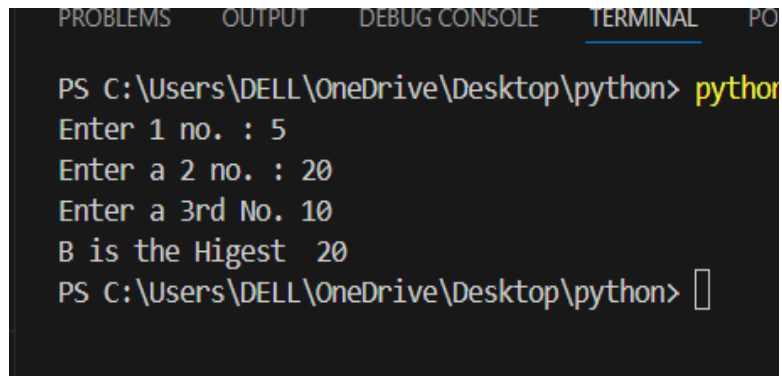
 print("Not Palindrome")

```
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\10.py"
Enter a 6-Digit no. : 123321
Palindrome
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\10.py"
Enter a 6-Digit no. : 145623
Not Palindrome
PS C:\Users\DELL\OneDrive\Desktop\python>
```

11. Wap in 'Python' language to accept three numbers, check and display the highest value on the # standard output device, using nested if ..else.

```
a=int(input("Enter 1 no. : "))
b=int(input("Enter a 2 no. : "))
c=int(input("Enter a 3rd No. "))

if a>b:
    if a>c:
        print("A is the highest",a)
    else:
        print("C is the highest ",c)
elif b>c:
    print("B is the Higest ",b)
else:
    print("C is the Higest ",c)
```



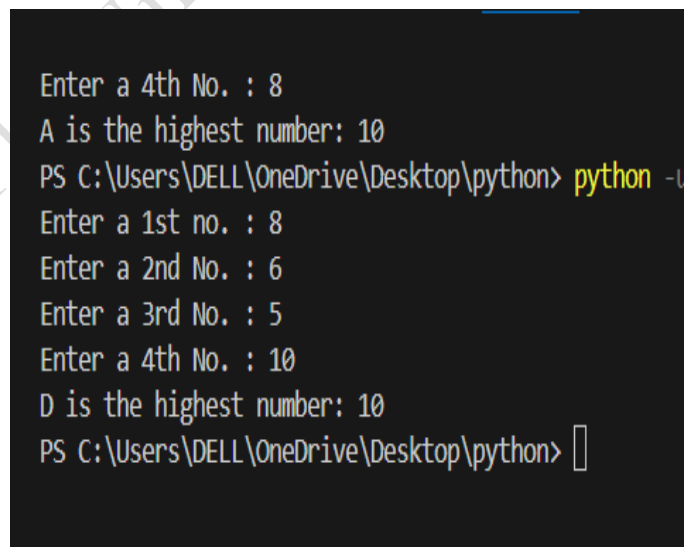
The screenshot shows a terminal window with the following text:

```
PS C:\Users\DELL\OneDrive\Desktop\python> python
Enter 1 no. : 5
Enter a 2 no. : 20
Enter a 3rd No. 10
B is the Higest  20
PS C:\Users\DELL\OneDrive\Desktop\python> 
```

12. Wap in 'Python' language to accept four numbers, check and display the highest value on the ## standard output device, using nested if..else.

```
a=int(input("Enter a 1st no. : "))
b=int(input("Enter a 2nd No. : "))
c=int(input("Enter a 3rd No. : "))
d=int(input("Enter a 4th No. : "))

if a > b:
    if a > c:
        if a > d:
            print("A is the highest number:", a)
        else:
            print("D is the highest number:", d)
    else:
        if c > d:
            print("C is the highest number:", c)
        else:
            print("D is the highest number:", d)
else:
    if b > c:
        if b > d:
            print("B is the highest number:", b)
        else:
            print("D is the highest number:", d)
    else:
        if c > d:
            print("C is the highest number:", c)
        else:
            print("D is the highest number:", d)
```



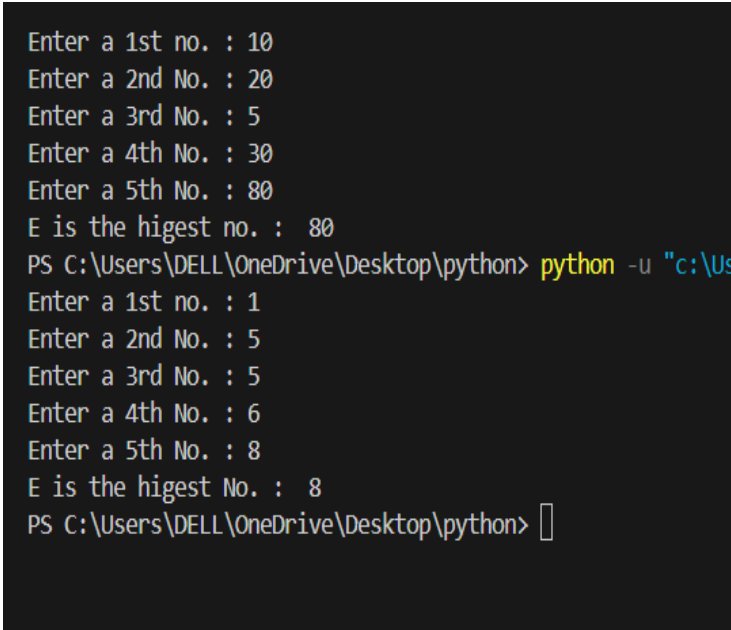
The screenshot shows a terminal window with the following text:

```
Enter a 4th No. : 8
A is the highest number: 10
PS C:\Users\DELL\OneDrive\Desktop\python> python -u
Enter a 1st no. : 8
Enter a 2nd No. : 6
Enter a 3rd No. : 5
Enter a 4th No. : 10
D is the highest number: 10
PS C:\Users\DELL\OneDrive\Desktop\python> 
```

13. Wap in 'Python' language to accept five numbers, check and display the highest value on the # standard output device, using nested if..else.

```
a=int(input("Enter a 1st no. : "))
b=int(input("Enter a 2nd No. : "))
c=int(input("Enter a 3rd No. : "))
d=int(input("Enter a 4th No. : "))
e=int(input("Enter a 5th No. : "))

if a > b:
    if a > c:
        if a > d:
            if a>e:
                print("A is the Higest No. : ",a)
            else:
                print("E is the Higest No. : ",e)
        else:
            if d>e:
                print("D is the highest No. :",d)
            else:
                print("E is the highest No. : ",e)
    else:
        if c>e:
            print("C is the highest No. : ",c)
        else:
            print("E is The Higest No. :",e)
else:
    if b>c:
        if b>d:
            if b>e:
                print("B is the Higest No. : ",b)
            else:
                print("E is the highest No. : ",e)
        else:
            if d>e:
                print("D is the highest No. : ",d)
            else:
                print("E is the highest no. : ",e)
    else:
        if c>e:
            print("C is the Higest No. : ",c)
        else:
            print("E is the highest No. : ",e)
```



The screenshot shows a terminal window with a black background and white text. It displays the execution of a Python program. The user enters five numbers: 10, 20, 5, 30, and 80. The program outputs "E is the highest no. : 80". Below this, the user runs the command `python -u "c:\Us` in the PowerShell prompt. The terminal then shows the user entering five new numbers: 1, 5, 5, 6, and 8. The program outputs "E is the highest No. : 8". The PowerShell prompt `PS C:\Users\DELL\OneDrive\Desktop\python>` is visible at the bottom.

```
Enter a 1st no. : 10
Enter a 2nd No. : 20
Enter a 3rd No. : 5
Enter a 4th No. : 30
Enter a 5th No. : 80
E is the highest no. : 80
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Us
Enter a 1st no. : 1
Enter a 2nd No. : 5
Enter a 3rd No. : 5
Enter a 4th No. : 6
Enter a 5th No. : 8
E is the highest No. : 8
PS C:\Users\DELL\OneDrive\Desktop\python> 
```

14. Wap in 'Python' language for an institute to accept a student's registration number and marks
obtained in the papers c, cpp and java. Display the marks detail in the following format as criteria
given below: -

- # • Each paper is of maximum 60 marks.
- # • Minimum pass marks in each paper is 30%.
- # • No any negative marking in any paper.
- # • The result will be PASS, if all the papers passed otherwise FAIL.

```
stud_reg = input("Enter a Regersation No. : " )
c=float(input("Enter a 'c' Marks (eg: maximum 60 marks) : "))
cpp =float(input("Enter a 'cpp' Marks (eg: maximum 60 marks) : "))
java = float(input("Enter a 'java' Marks (eg: maximum 60 marks) : "))
```

```
max_marks= 60
```

```
pass_marks=0.3* max_marks
```

```
result = "FAIL"
```

```
if c<0 or c>max_marks:
```

```
    print("C Marks is Invalid",c)
```

```
    result = "INVALID"
```

```
else:
```

```
    if cpp<0 or cpp>max_marks:
```

```
        print("CPP Marks is Invalid. ",cpp)
```

```
        result = "INVALID"
```

```
    else:
```

```
        if java < 0 or java > max_marks:
```

```
            print("Java Marks is Invalid",java)
```

```
            result = "INVALID"
```

```
        else:
```

```
            # Nested PASS/FAIL checking
```

```
            if c >= pass_marks:
```

```
                if cpp >= pass_marks:
```

```
                    if java >= pass_marks:
```

```
                        result = "PASS"
```

```
                    else:
```

```
                        result = "FAIL"
```

```
                else:
```

```
                    result = "FAIL"
```

```
            else:
```

```
                result = "FAIL"
```

```
print("=====")
```

```
print("                Marks Detail")
```

```
print("=====")
```

```
print(" Registration Number  :: ",stud_reg)
```

```
print(" Marks Obtained in C   :: ",c)
```

```
print(" Marks Obtained in Cpp :: ",cpp)
```

```
print(" Marks Obtained in Java :: ",java)
```

```
print(" Result remarks       :: ",result)
```

```
print("=====")
```

```
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\14.py"
Enter a Regersation No. : Ask45
Enter a 'c' Marks (eg: maximum 60 marks) : 50
Enter a 'cpp' Marks (eg: maximum 60 marks) : 60
Enter a 'java' Marks (eg: maximum 60 marks) : -06
Java Marks is Invalid -6.0
=====
                Marks Detail
=====
Registration Number      :: Ask45
Marks Obtained in C     :: 50.0
Marks Obtained in Cpp   :: 60.0
Marks Obtained in Java  :: -6.0
Result remarks          :: INVALID
=====
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\python\14.py"
Enter a Regersation No. : 45
Enter a 'c' Marks (eg: maximum 60 marks) : 60
Enter a 'cpp' Marks (eg: maximum 60 marks) : 60
Enter a 'java' Marks (eg: maximum 60 marks) : 50
=====
                Marks Detail
=====
Registration Number      :: 45
Marks Obtained in C     :: 60.0
Marks Obtained in Cpp   :: 60.0
Marks Obtained in Java  :: 50.0
Result remarks          :: PASS
=====
PS C:\Users\DELL\OneDrive\Desktop\python>
```

15.Wap in 'C' language for an organization to accept employee number and basic salary, calculate
 # DA@10% of basic salary if the basic salary is more than 4000 otherwise @15%. Similarly calculate
 # HRA@20% of basic salary if the basic salary is less than 8000 otherwise @15%. Calculate and display
 # the salary detail of the employee in the following format, as income tax to be deducted @12% of
 Gross
 # Salary if Gross Salary exceeds from 12000.

```

eno = input("Enter a Employees No. : ")
bas_saly = float(input("Enetr a Basics Salary. : "))
if bas_saly>4000:
    da=bas_saly*10/100
else:
    da=bas_saly*15/100
if bas_saly<8000:
    hra =bas_saly*20/100
else:
    hra= bas_saly*15/100
gross=bas_saly+da+hra
if gross>12000:
    itax=gross*12/100
else:
    itax=0
print("=====")
print("                Salary Detail")
print("=====")
print("Employee Number      : ",eno)
print("Basic Salary in Rs.   : ",bas_saly)
print("DA in Rs.              : ",da)
print("HRA in Rs.             : ",hra)
print("Gross Salary in Rs.    : ",gross)
print("Income Tax Deduction in Rs. : ",itax)
print("Net Payable Amount in Rs. : ",gross-itax)
print("=====")
  
```

```

PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\p
Enter a Employees No. : Rays45
Enetr a Basics Salary. : 10000
=====
                Salary Detail
=====
Employee Number      :    Rays45
Basic Salary in Rs.   :   10000.0
DA in Rs.             :    1000.0
HRA in Rs.            :    1500.0
Gross Salary in Rs.   :   12500.0
Income Tax Deduction in Rs. :   1500.0
Net Payable Amount in Rs. :   11000.0
=====
PS C:\Users\DELL\OneDrive\Desktop\python> python -u "c:\Users\DELL\OneDrive\Desktop\p
Enter a Employees No. : RAYS85
Enetr a Basics Salary. : 50000
=====
                Salary Detail
=====
Employee Number      :    RAYS85
Basic Salary in Rs.   :   50000.0
DA in Rs.             :    5000.0
HRA in Rs.            :    7500.0
Gross Salary in Rs.   :   62500.0
Income Tax Deduction in Rs. :    7500.0
Net Payable Amount in Rs. :   55000.0
=====
PS C:\Users\DELL\OneDrive\Desktop\python> 
  
```